

Learning goals: Linux OS commands **apropos**, file permissions

NOTE: Save screenshots of each exercise, and upload your work to your github account as Lab7.pdf by end of Thursday Aug 6 using git commands. Also upload a screenshot of your git commands in your Lab7 directory.

EXERCISES Start a fresh terminal for this Lab.

- (1) The **apropos** command allows you to search the system's manual pages for commands that match a specific task or keyword. This is particularly useful when you need to find a command that performs a specific task but you don't know the name of the command. Try the following and display the output in your lab document.

- (i) **apropos "search for files"**
- (ii) **apropos "list directory contents"**. Another way to get the same output is **apropos "^list.*directory"** (note the use of wildcards here). The **^** denotes that there is no other character before the word 'list' in the search.
- (iii) **apropos "file compress"**
- (iv) **apropos "file.*copy\$"** (The **\$** denotes that there is no character to be searched after the word 'copy'.

Come up with 3 more keywords/phrases that you can try for this command and got meaningful output (show the screenshots of all your outputs)

- (2) File permissions. Try the following exercises and explain the output in your work.
 - (i) **touch newfile** followed by **ls -l newfile**. Describe the permissions set for the user, group and everyone else.
 - (ii) Print the output of **umask** command. Explain what is the actual level of default permission based on this output for a file, and for a directory
 - (iii) Create a new file in your Lab7 directory and print and explain the current permissions for the file. Write down the octal values for the user, group and everyone else
 - (iv) Create a new directory in your Lab7 directory and print and and explain the current permissions for the file. Write down the octal values for the user, group and everyone else.
 - (v) Copy the new file from above to the new directory and list the permissions of the file inside the directory. Is it the same as the file outside the directory? What does this tell you about the copying action?
 - (vi) Change the permissions of the directory such that it is only readable and writable by the user and the group, and only readable by everyone else. What happens to the permissions of the file inside the directory? Does it inherit the new permissions you just set? What are the new octal values for the directory permissions?
 - (vii) Move the above-created new file to the new directory under a different name. Does the moving action change the file permissions?